Groups and Coding

Groups and Coding

Coding of Binary Information and Error Detection

Decoding and Error Correction

Coding of Binary Information and Error Detection

Encoding function

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Coding theory

In today's modern world of communication, data items are constantly being transmitted from point to point. The basic problem in transmission of data is that of receiving the data as sent and not receiving a distorted piece of data.

Coding theory has developed techniques for introducing redundant information(引入冗余信息) *in transmitted data that help in detecting, and sometimes in correcting, errors.*

Unit of information

Message(消息) is a finite sequence of characters from a finite alphabet $B = \{0, I\}$

Word(字) is a sequence of m 0's and I 's.

Groups

The set B is a group under the binary operation + (mod 2 addition)

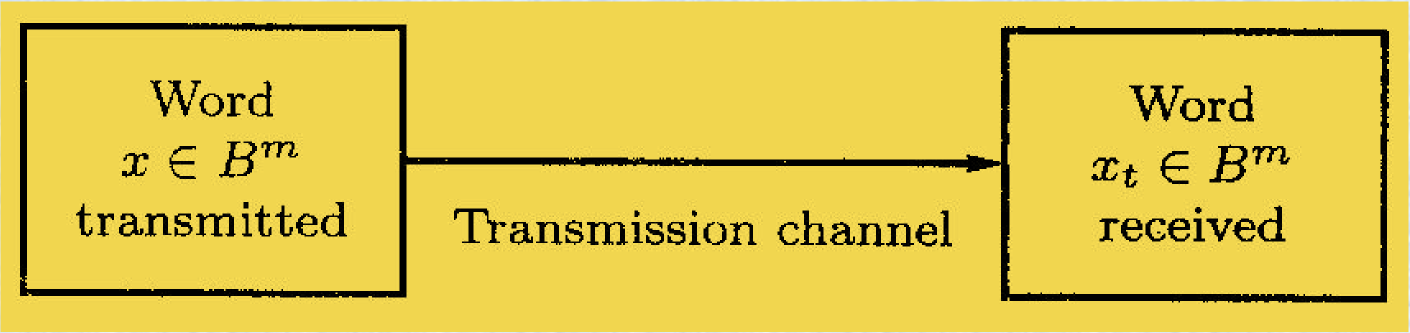
It follows that $B^m = B \times B \times \ldots \times B$ (n factors) is a group under the operator $\oplus$ defined by

$$(x_1, x_2, \ldots, x_m) \oplus (y_1, y_2, \ldots, y_m)$$
$$= (x_1 + y_1, x_2 + y_2, \ldots, x_m + y_m)$$

An element in B^m will be written as $(b_1, b_2, \ldots, b_m)$ or more simply as $b_1 b_2 \ldots b_m$

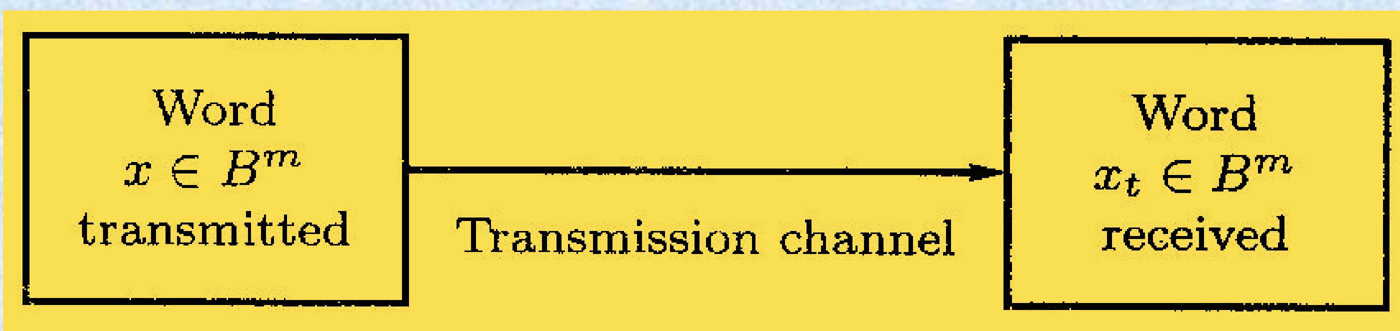
Transmission channel and Noise

An element $x \in B^m$ is sent through the transmission channel and is received as an element $x_t \in B^m$.



Transmission channel and Noise

Noise(噪声) in the transmission channel may cause a 0 to be received as a l, or vice versa, lead $x \neq x_t$



对

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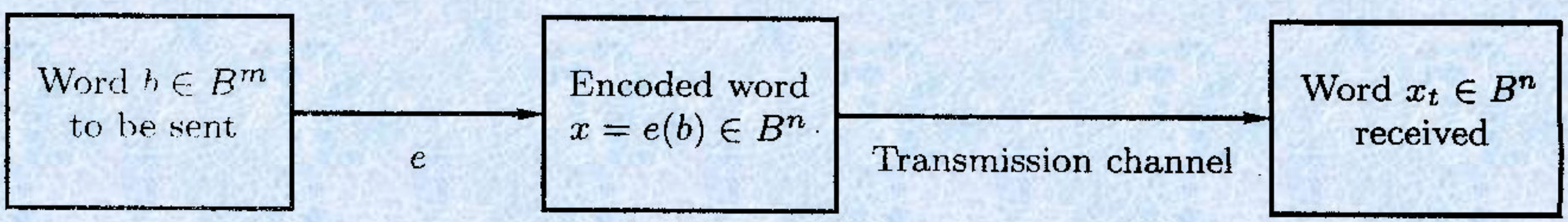
Encoding function - 编码函数

Choose: an integer $n>m$ and a one-to-one function e: $B^m \to B^n$

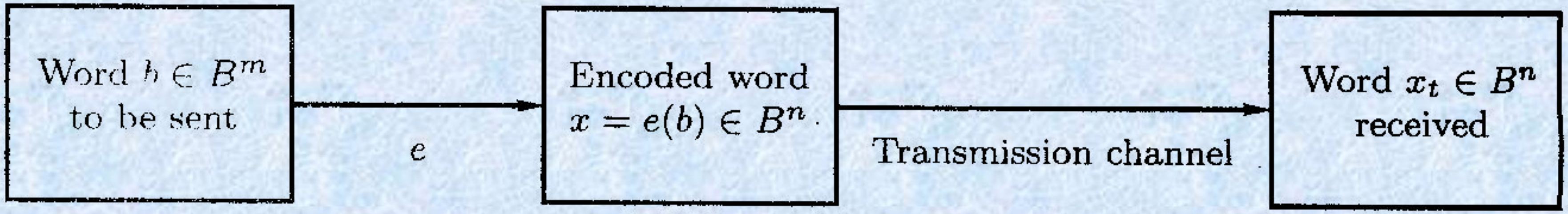
- e is called an (m, n) *encoding function*, representing every word in B^m as a word in B^n.
- If $b \in B^m$, then $e(b)$ is called the *code word*(码字) representing b.



Encoding function

If the transmission channel is noiseless, then $x_t = x$ for all x in B^n.

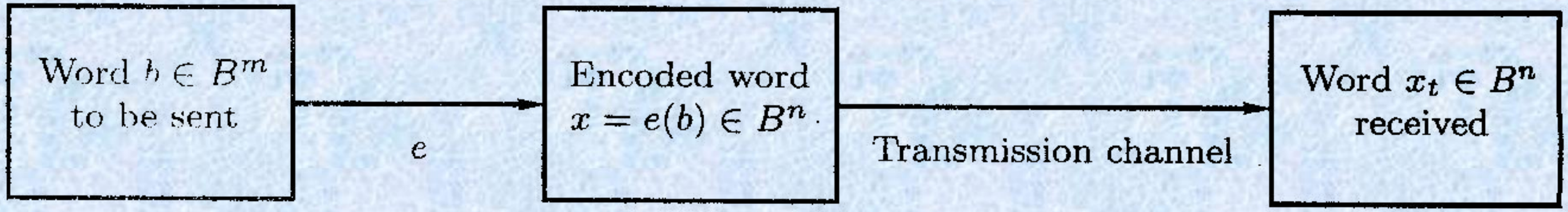
In this case $x = e(b)$ is received for each $b \in B^m$, and since e is a known function, b may be identified.



Encoding function

In general, errors in transmission do occur.

We will say that the code word $x = e(b)$ has been transmitted with *k or fewer errors* if x and x_t differ in at least 1 but no more than k positions.



Error detect – 差错检测

Let e: $B^m \to B^n$ be an (m, n) encoding function.

e *detects k or fewer errors* if whenever $x = e(b)$ is transmitted with k or fewer errors, then x_t is not a code word (thus x_t could not be x and therefore could not have been correctly transmitted).

For $b \in B^n$, the number of I's in x is called the weight(权) of x and is denoted by $|x|$

Example 1

Find the weight of each of the following words in B^5.

x=01000	$\|x\|$=1
x=11100	$\|x\|$=3
x=00000	$\|x\|$=0
x=11111	$\|x\|$=5

Example 2
parity check code –奇偶校验码

The following encoding function e: $B^m \to B^{m+1}$ is called the *parity (m, m+1) check code*: If $b = b_1b_2\ldots b_m \in B^m$, define

$$e(b) = b_1b_2\ldots b_m\, b_{m+1}$$

where

$$b_{m+1} = \begin{cases} 0 & \text{if } |b| \text{ is even} \\ 1 & \text{if } |b| \text{ is odd.} \end{cases}$$

Example 2
parity check code

Let $m = 3$. Then

$e(000) = 0000$

$e(001) = 0011$

$e(010) = 0101$

$e(011) = 0110$

$e(100) = 1001$

$e(101) = 1010$

$e(110) = 1100$

$e(111) = 1111$

Suppose now that $b = 111$. Then $x = e(b) = 1111$

Example 3

Consider the (m, $3m$) encoding function e: $B^m \to B^{3m}$.

If $b = b_1b_2\ldots b_m \in B^m$, define

$$e(b) = b_1b_2\ldots b_mb_1b_2\ldots b_mb_1b_2\ldots b_m$$

Example 3

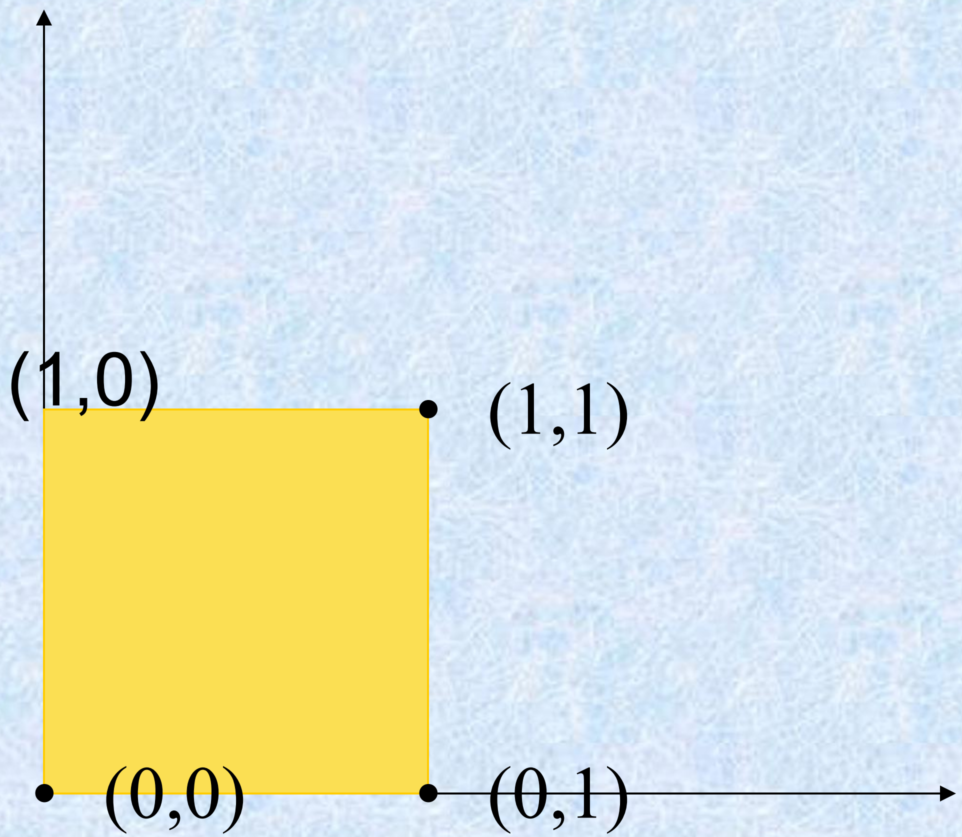
$$\left.\begin{array}{l} e(000) = 000000000 \\ e(001) = 001001001 \\ e(010) = 010010010 \\ e(011) = 011011011 \\ \hline e(100) = 100100100 \\ e(101) = 101101101 \\ e(110) = 110110110 \\ e(111) = 111111111 \end{array}\right\} \text{code word}$$

Suppose now that *b* = 011

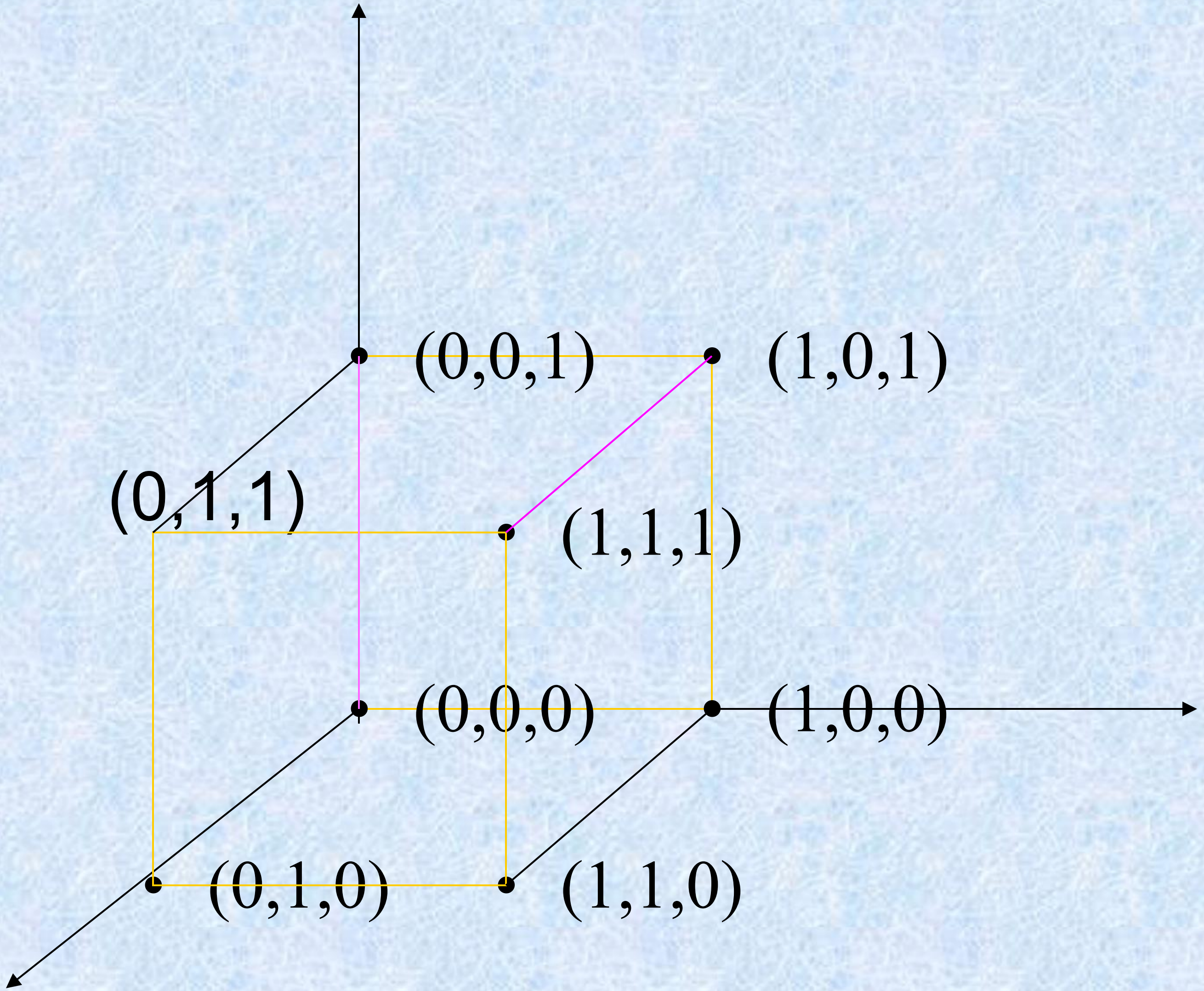
Then *e*(011) = 011011011.

Assume now we receive the word 011111011. This is not a code word, so we have detected the error.

(00,10,01,11)
(00,11)



$B^m(0,1) \to B^n(000,111)$



Hamming distance – 海明距离

Let x and y be words in B^m. The *Hamming distance* $\delta(x, y)$ between x and y is the weight, $|x \oplus y|$, of $x \oplus y$.

The distance between $x = x_1x_2\ldots x_m$ and $y = y_1y_2\ldots y_m$ is the number of various of i such that $x_i \neq y_i$, that is, the number of positions in which x and y differ.

Using the weight of $x \oplus y$ is a convenient way to count the number of different positions.

Example 4

Find the distance between x and y:

(a) $x = 110110$, $y = 000101$

(b) $x = 001100$, $y = 010110$.

Solution

(a) $x \oplus y = 110011$, so $| x \oplus y | = 4$

(b) $x \oplus y = 011010$, so $| x \oplus y | = 3$

Theorem 1
properties of distance function

Let x, y, and z be elements of B^m. Then

(a) $\delta(x, y) = \delta(y, x)$

(b) $\delta(x, y) \geq 0$

(c) $\delta(x, y) = 0$ if and only if $x = y$

(d) $\delta(x, y) \leq \delta(x, z) + \delta(z, y)$

Proof of (d)

$$\delta(x, y) = |\, x \oplus y \,| = |\, x \oplus \mathbf{0} \oplus y \,|$$
$$= |\, x \oplus z \oplus z \oplus y \,|$$
$$\leq |\, x \oplus z \,| + |\, z \oplus y \,|$$

Minimum distance

The *minimum distance* of an encoding function $e\colon B^m \to B^n$ is the minimum of the distances between all distinct pairs of code words; that is,

$$\min\{\delta(e(x), e(y)) \mid x, y \in B^m\}$$

Example 5

Consider the following (2, 5) encoding function e:

$$\left.\begin{aligned} e(00) &= 00000 \\ e(01) &= 00111 \\ e(10) &= 01110 \\ \underline{e(11)} &\underline{= 11111} \end{aligned}\right\} \text{code word}$$

Theorem 2

An (m, n) encoding function $e: B^m \to B^n$ can detect k or fewer errors

if and only if

its minimum distance is at least k + l.

Proof

$\Leftarrow$ the minimum distance is at least $k + 1$

Let $b \in B^m$, and let $x = e(b) \in B^n$ be the code word representing b.

- x is transmitted and is received as x_t. If x_t were a code word different from x, then $\delta(x, x_t) \geq k+l$, so x would be transmitted with k + l or more errors.
- Thus, if x is transmitted with k or fewer errors, then x_t cannot be a code word.
- This means that e can detect k or fewer errors.

Proof

e can detect k or fewer errors $\Rightarrow$

Suppose that the minimum distance between code words is $r \leq k$

Let x and y be code words with $\delta(x, y) = r$.

- If $x = y$, that is, if x is transmitted and is mistakenly received as y, then $r \leq k$ errors have been committed and have not been detected.

Thus it contradict with e can detect k or fewer errors.

Q.E.D.

Example 6

How many errors will *e* detect?

Consider the (3, 8) encoding function e: $B^3 \to B^8$ defined by

$$\left.\begin{array}{l}
e(000) = 00000000 \\
e(001) = 10011100 \\
e(010) = 00101101 \\
e(011) = 10010101 \\
\hline
e(100) = 10100100 \\
e(101) = 10001001 \\
e(110) = 00011100 \\
e(111) = 00110001
\end{array}\right\} \text{code word}$$

Group Codes – 群码

An (m, n) encoding function $e: B^m \to B^n$ is called a *group code* if

$e(B^m) = \{e(b) \mid e(b) \in B^n\} = \mathrm{Ran}(e)$

is a subgroup of B^n

Subgroups

Recall from the definition of subgroup give in Section 9.4 that N is a subgroup of B^n if

(a) the identity of B^n is in N,

(b) if x and y belong to N, then $x \oplus y \in N$, and

(c) if x is in N, then its inverse is in N.

Example 7: is e a group code?

Consider the (3, 6) encoding function e: $B^3 \to B^6$ defined by

$$\left.\begin{array}{l}
e(000) = 000000 \\
e(001) = 001100 \\
e(010) = 010011 \\
e(011) = 011111 \\
\hline
e(100) = 100101 \\
e(101) = 101001 \\
e(110) = 110110 \\
e(111) = 111010
\end{array}\right\} \text{code word}$$

Example 7: is e a group code?

We must show that the set of all code words

N = {000000, 001100, 010011, 011111, 100101, 101001, 110110, 111010}

is a subgroup of B^6.

Theorem 3

Let e: $B^m \rightarrow B^n$ be a group code. The minimum distance of e is the minimum weight of a nonzero code word.

Proof of Theorem 3

Let δ be the minimum distance of the group code, and suppose that $\delta = \delta(x, y)$, where x and y are distinct code words.

Also, let η be the minimum weight of a nonzero code word and suppose that $\eta = |z|$ for a code word z.

Proof of Theorem 3

Since e is a group code, $x \oplus y$ is a nonzero code word. Thus

$\delta = \delta(x, y) = |x \oplus y| \geq \eta$.

On the other hand, since $\mathbf{0}$ and z are distinct code words,

$\eta = |z| = |z \oplus \mathbf{0}| = \delta(z, \mathbf{0}) \geq \delta$

Hence $\eta = \delta$.

Q.E.D.

Example 8

The minimum distance of the group code in

$$\left.\begin{array}{l} e(000)=000000 \\ e(001)=001100 \\ e(010)=010011 \\ e(011)=011111 \\ \hline e(100)=100101 \\ e(101)=101001 \\ e(110)=110110 \\ e(111)=111010 \end{array}\right\} \text{code word}$$

is 2

To check this directly would require 28 different calculations.

Constructing

group code

A review on Boolean matrices

mod-2 sum D ⊕ E

mod-2 Boolean product D * E

Theorem 4

Let **D** and **E** be $m \times p$ Boolean matrices, and let **F** be a $p \times n$ Boolean matrix. Then

(**D** ⊕ **E**) * **F** = (**D** * **F**) ⊕ (**E** * **F**).

That is, a distributive property holds for ⊕ and *.

Example 5

Let $B = \{0, I\}$, and let + be the operation defined on B as follows:

+	0	1
0	0	1
1	1	0

Then B is a group.

Let $\mathbf{D} = [\, d_{ij} \,]$ be an $m \times p$ Boolean matrix, and let $\mathbf{E} = [\, e_{ij} \,]$ be a $p \times n$ Boolean matrix. We define the **mod-2 Boolean product $\mathbf{D} * \mathbf{E}$** as the $m \times n$ matrix $\mathbf{F} = [\, f_{ij} \,]$, where

$$f_{ij} = d_{i1} \cdot e_{1j} + d_{i2} \cdot e_{2j} + \cdots + d_{ip} \cdot e_{pj}, \qquad 1 \le i \le m, \quad 1 \le j \le n.$$

This type of multiplication is illustrated in Figure 11.3. Compare this with similar figures in Section 1.5.

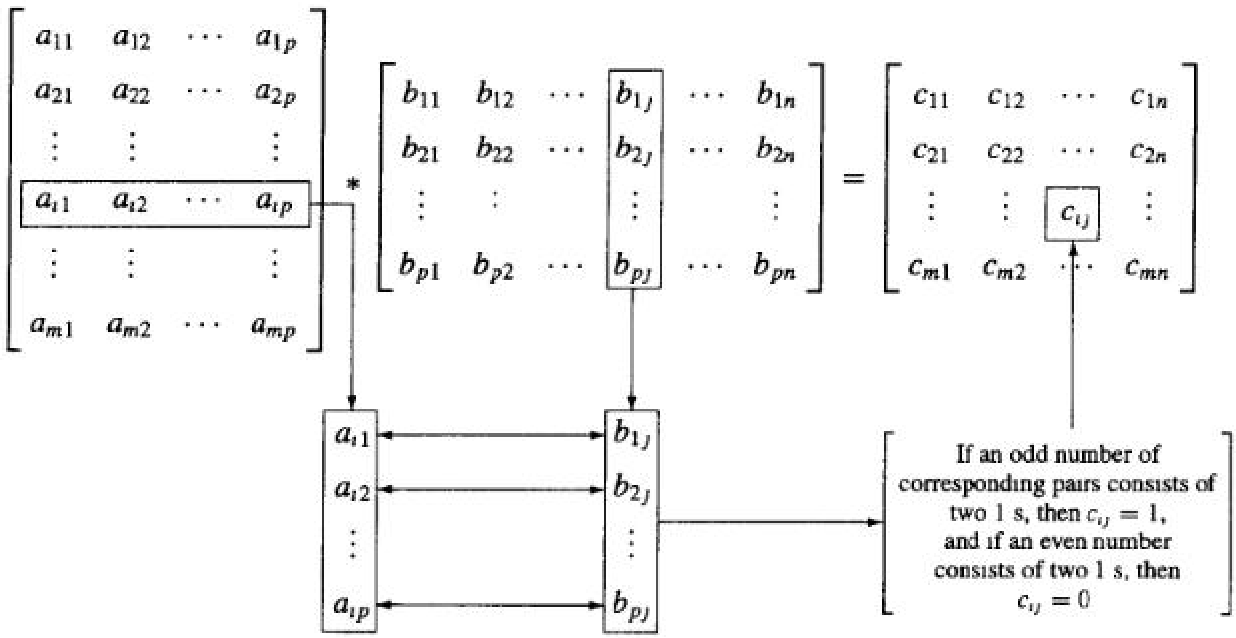


Figure 11.3

EXAMPLE 10 We have

$$\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 \cdot 1 + 1 \cdot 1 + 0 \cdot 0 & 1 \cdot 0 + 1 \cdot 1 + 0 \cdot 1 \\ 0 \cdot 1 + 1 \cdot 1 + 1 \cdot 0 & 0 \cdot 0 + 1 \cdot 1 + 1 \cdot 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \quad \blacksquare$$

Boolean Products

Let **A**=[a_{ij}] be an $m \times k$ zero-one matrix, & let **B**=[b_{ij}] be a $k \times n$ zero-one matrix,

The *boolean product* of **A** and **B** is like normal matrix ×, but using ∨ instead of + in the row-column "vector dot product":

$$\mathbf{A} \odot \mathbf{B} = \mathbf{C} = [c_{ij}] = \left[\bigvee_{\ell=1}^{k} a_{i\ell} \wedge b_{\ell j} \right]$$

EXAMPLE 8 Find the Boolean product of $\mathbf{A}$ and $\mathbf{B}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}, \qquad \mathbf{B} = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}.$$

Solution: The Boolean product $\mathbf{A} \odot \mathbf{B}$ is given by

$$\mathbf{A} \odot \mathbf{B} = \begin{bmatrix} (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \\ (0 \wedge 1) \vee (1 \wedge 0) & (0 \wedge 1) \vee (1 \wedge 1) & (0 \wedge 0) \vee (1 \wedge 1) \\ (1 \wedge 1) \vee (0 \wedge 0) & (1 \wedge 1) \vee (0 \wedge 1) & (1 \wedge 0) \vee (0 \wedge 1) \end{bmatrix}$$

$$= \begin{bmatrix} 1 \vee 0 & 1 \vee 0 & 0 \vee 0 \\ 0 \vee 0 & 0 \vee 1 & 0 \vee 1 \\ 1 \vee 0 & 1 \vee 0 & 0 \vee 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}.$$

Notation

We shall now consider the element

$x = x_1 x_2 \ldots x_n \in B^n$

as the $1 \times n$ matrix

$[x_1\ x_2\ \ldots\ x_n\ ]$

Theorem 5

Let m and n be nonnegative integers with $m < n$, $r = n - m$, and let $\mathbf{H}$ be an $n \times r$ Boolean matrix.

Then the function $f_H: B^n \to B^r$ defined by

$$f_H(x) = x * \mathbf{H}, \quad x \in B^n;$$

is a homomorphism from the group B^n to the group B^r.

Proof

If x and y are elements in B^n, then

$f_H(x \oplus y) = (x \oplus y) * \mathbf{H}$

$= (x * \mathbf{H}) \oplus (y * \mathbf{H})$ by Theorem 4

$= f_H(x) \oplus f_H(y)$.

Hence f_H is a homomorphism from B^n to B^r

Q.E.D.

Corollary 1

Let *m, n, r*, **H**, and f_H be as in Theorem 5. Then

$N = \{x \in B^n \mid x * \mathbf{H} = \mathbf{0}\}$

is a normal subgroup of B^n.

Proof:

N is the kernel of the homomorphism f_H, so it is a normal subgroup of B^n.

Parity check matrix – 一致性检验矩阵

Let $m < n$ and $r = n - m$, the following $n \times r$ Boolean matrix is called a *parity check matrix*.

$$\mathrm{H} = \begin{bmatrix} \mathrm{H}_{m\times r} \\ \mathrm{I}_r \end{bmatrix} = \left[\begin{array}{cccc} h_{11} & h_{12} & \cdots & h_{1r} \\ h_{21} & h_{22} & \cdots & h_{2r} \\ \vdots & \vdots & & \vdots \\ h_{m1} & h_{m2} & \cdots & h_{mr} \\ \hline 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{array}\right]$$

Encoding function

Define an encoding function e_H: $B^m \rightarrow B^n$

If $b = b_1 b_2 \ldots b_m$,

let $x = e_H(b) = b_1 b_2 \ldots b_m\, x_1 x_2 \ldots x_r$, where

$$
\begin{aligned}
x_1 &= b_1 \cdot h_{11} + b_2 \cdot h_{21} + \ldots + b_m \cdot h_{m1} \\
x_2 &= b_1 \cdot h_{12} + b_2 \cdot h_{22} + \ldots + b_m \cdot h_{m2} \\
&\vdots \\
x_r &= b_1 \cdot h_{1r} + b_2 \cdot h_{2r} + \ldots + b_m \cdot h_{mr}
\end{aligned}
\qquad (1)
$$

e_H: $B^m \to B^n$ in matrix format

$$e_{\mathrm{H}}(B^m) = B^m * \begin{bmatrix} \mathrm{I}_m & \mathrm{H}_{m\times r} \end{bmatrix}$$

$$= \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1m} \\ b_{21} & b_{22} & \cdots & b_{2m} \\ \vdots & \vdots & & \vdots \\ b_{2^m 1} & b_{2^m 2} & \cdots & b_{2^m m} \end{bmatrix} \begin{bmatrix} 1 & 0 & \cdots & 0 & h_{11} & h_{12} & \cdots & h_{1r} \\ 0 & 1 & \cdots & 0 & h_{21} & h_{22} & \cdots & h_{2r} \\ \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 & h_{m1} & h_{m2} & \cdots & h_{mr} \end{bmatrix}$$

$$= \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1m} & x_{11} & x_{12} & \cdots & x_{1r} \\ b_{21} & b_{22} & \cdots & b_{2m} & x_{21} & x_{22} & \cdots & x_{2r} \\ \vdots & \vdots & & \vdots & \vdots & \vdots & & \vdots \\ b_{2^m 1} & b_{2^m 2} & \cdots & b_{2^m m} & x_{2^m 1} & x_{2^m 2} & \cdots & x_{2^m r} \end{bmatrix}$$

Theorem 6

Let $x = y_1 y_2 \ldots y_m x_1 \ldots x_r \in B^n$. Then

$x * \mathbf{H} = \mathbf{0}$

if and only if

$x = e_H(b)$ for some $b \in B^m$.

Proof: $x * \mathbf{H} = \mathbf{0} \Rightarrow x = e_H(b)$ for some $b \in B^m$

Suppose that x * **H = 0**

$$y_1 \cdot h_{11} + y_2 \cdot h_{21} + \ldots + y_m \cdot h_{m1} + x_1 = 0$$

$$y_1 \cdot h_{12} + y_2 \cdot h_{22} + \ldots + y_m \cdot h_{m2} + x_2 = 0$$

$$\vdots$$

$$y_1 \cdot h_{1r} + y_2 \cdot h_{2r} + \ldots + y_m \cdot h_{mr} + x_r = 0$$

$$\mathrm{H} = \begin{bmatrix} \mathrm{H}_{m \times r} \\ \mathrm{I}_r \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & \cdots & h_{1r} \\ h_{21} & h_{22} & \cdots & h_{2r} \\ \vdots & \vdots & & \vdots \\ h_{m1} & h_{m2} & \ldots & h_{mr} \\ \hline 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

$x = y_1 y_2 \ldots y_m \, x_1 x_2 \ldots x_r$

Proof: $x * \mathbf{H} = \mathbf{0} \Rightarrow x = e_H(b)$ for some $b \in B^m$

Note that $x_i+x_i=0$. So add x_i to i^{th} row and get

$$x_i = y_1 \cdot h_{1i} + y_2 \cdot h_{2i} + \ldots + y_m \cdot h_{mi}$$

Letting $b_1 = y_1$, $b_2 = y_2$,..., $b_m = y_m$, we see that $x_1, x_2, \ldots, x_r$ satisfy the equations in (I).

Thus $b = b_1b_2\ldots b_m \in B^m$ and $x = e_H(b)$

$x = e_H(b) = b_1b_2\ldots b_m\, x_1x_2\ldots x_r$

$$\begin{aligned}
x_1 &= b_1 \cdot h_{11} + b_2 \cdot h_{21} + \ldots + b_m \cdot h_{m1} \\
x_2 &= b_1 \cdot h_{12} + b_2 \cdot h_{22} + \ldots + b_m \cdot h_{m2} \\
&\vdots \\
x_r &= b_1 \cdot h_{1r} + b_2 \cdot h_{2r} + \ldots + b_m \cdot h_{mr}
\end{aligned}$$

Proof: $x * \mathbf{H} = \mathbf{0} \Leftarrow x = e_H(b)$ for some $b \in B^m$

Conversely if $x = e_H(b)$, the equations in (I) can be rewritten by adding x_i to both sides of the i^{th} equation, i = l, 2,..., n, as

$$
\begin{aligned}
&b_1 \cdot h_{11} + b_2 \cdot h_{21} + \ldots + b_m \cdot h_{m1} + x_1 = 0 \\
&b_1 \cdot h_{12} + b_2 \cdot h_{22} + \ldots + b_m \cdot h_{m2} + x_2 = 0 \\
&\vdots \\
&b_1 \cdot h_{1r} + b_2 \cdot h_{2r} + \ldots + b_m \cdot h_{mr} + x_r = 0
\end{aligned}
$$

which shows $x * \mathbf{H} = \mathbf{0}$

Q.E.D.

Corollary 2

$e_H(B^m) = \{e_H(b) \mid b \in B^m\}$ is a subgroup of B^n

Proof :

The result follows from the observation that

$e_H(B^m) = \ker(f_H)$

and from Corollary I.

Thus e_H is a group code.

Example 11

Let m = 2, n = 5, and

$$H = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Determine the group code e_H: $B^2 \to B^5$.

Solution 11

$$e(B^m) = B^m * \mathrm{H}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 1 \\ 1 & 0 \\ 1 & 1 \end{bmatrix} * \begin{bmatrix} 1 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 0 & 1 \end{bmatrix}$$

Homework

19,23,25,27   P412

Thanks for your attention!

Coming up next …

Decoding and Error Correction

练习

Let

$$\mathbf{H} = \begin{bmatrix} 0 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

be a parity check matrix. Determine the (2, 5) group code function $e_H : B^2 \to B^5$.

Decoding and Error Correction

译码与纠错

$$\boxed{\begin{array}{c}\text{Word } b \in B^m\\ \text{to be sent}\end{array}} \xrightarrow{e} \boxed{\begin{array}{c}\text{Encoded word}\\ x = e(b) \in B^n.\end{array}} \xrightarrow{\text{Transmission channel}} \boxed{\begin{array}{c}\text{Word } x_t \in B^n\\ \text{received}\end{array}}$$

$$\begin{tikzcd}
B^n \arrow[rr, "f_H(x) = x*H"] \arrow[dr] & & B^r \arrow[dl, shift left] \\
& B^n/N \arrow[ur, shift left] &
\end{tikzcd}$$

$f_R(x) = [x]_R = x \oplus N$

xRy if only if $x^{-1} \oplus y \in N$

$f_H(x) = f_H(y)$

$\overline{F}(xN) = f_H(x) = x*H$

$\{xN \mid x \in B^n\} = \{(x*H = y \mid y \in B^r)\}$

$N = \{x = e(b), b \in B^m\} = \{(x*H = 0) \mid x \in B^n\}$

Decoding Function – 译码函数

Consider an (m, n) encoding function

$e: B^m \to B^n$.

Once the encoded word $x = e(b) \in B^n$, for $b \in B^m$, is received as the word x_t, we are faced with the problem of identifying the word b that was the original message.

Decoding Function

Consider an (m, n) encoding function

$e: B^m \to B^n.$

An onto function $d: B^n \to B^m$ is called an (n, m) *decoding function associated with e*(与e关联的译码函数) if $d(x_t) = b' \in B^m$ is such that when the transmission channel has no noise, then $b' = b$, that is,

$d \circ e = 1_{B^m}$

Decoding Function

The decoding function *d* is required to be onto so that every received word can be decoded to give a word in B^m.

It decodes properly received words correctly, but the decoding of improperly received words may or may not be correct.

Example 1

Consider the parity check code that is defined in Example 2 of Section 11.1

Define the decoding function d: $B^{m+1} \to B^m$.

If $y = y_l y_2 \ldots y_m y_{m+l} \in B^{m+1}$, then $d(y) = y_l y_2 \ldots y_m$

Observe that if $b = b_l b_2 \ldots b_m \in B^m$, then

$(d \circ e)(b) = d(e(b)) = b$

so $d \circ e = 1_{B^m}$

For a concrete example, let $m = 4$.

Then $d(10010) = 1001$ and $d(11001) = 1100$

Decoding function

Let e be an (m, n) encoding function and let d be an (n, m) decoding function associated with e.

The pair (e, d) is said to *correct k or fewer errors*

- if whenever $x = e(b)$ is transmitted correctly or with k or fewer errors and x_t is received, then $d(x_t) = b$. Thus x_t is decoded as the correct message b.

Example 2

Consider the (m, $3m$) encoding function defined in Example 3 of Section 11.1. Define the decoding function d: $B^m \rightarrow B^{3m}$.

Let $y = y_1 y_2 \ldots y_m y_{m+1} \ldots y_{2m} y_{2m+1} \ldots y_{3m}$, Then

$d(y) = z_1 z_2 \ldots z_m$

where

$$z_i = \begin{cases} 1 & \text{if } \{y_i, y_{i+m}, y_{i+2m}\} \text{ has at least two 1's} \\ 0 & \text{if } \{y_i, y_{i+m}, y_{i+2m}\} \text{ has less than two 1's} \end{cases}$$

E.g. x_t = 011011111, then $d(x_t)$ = 011

Maximum likelihood technique – 极大似然技术

Since B^m has 2^m elements, there are 2^m code words in B^n. List it as

$x^{(1)}, x^{(2)}, \ldots, x^{(2^m)}$

If the received word is x_t, we compute $\delta(x^{(i)}, x_t)$ for $1 \leq i \leq 2^m$, and choose the first code word, say it is $x^{(s)}$, such that

$$\min_{1 \leq i \leq 2^m} \{\delta(x^{(i)}, x_t)\} = \delta(x^{(s)}, x_t)$$

Maximum likelihood decoding function

That is, $x^{(s)}$ is a code word that is closest to x_t and the first in the list.

If $x^{(s)} = e(b)$, we define the *maximum likelihood decoding function* d associated with e by

$d(x_t) = b$

Theorem 1

Suppose that *e* is an (*m, n*) encoding function and *d* is a maximum likelihood decoding function associated with *e*. Then

(*e, d*) can correct *k* or fewer errors

if and only if

the minimum distance of *e* is at least 2*k* + l.

Example 3

How many errors can (e, d) correct?

The (3, 8) encoding function $e: B^3 \rightarrow B^8$

$$\left.\begin{array}{l}
e(000) = 00000000 \\
e(001) = 10011100 \\
e(010) = 00101101 \\
e(011) = 10010101 \\
\hline
e(100) = 10100100 \\
e(101) = 10001001 \\
e(110) = 00011100 \\
e(111) = 00110001
\end{array}\right\} \text{code word}$$

and let d be an (8, 3) maximum likelihood decoding function associated with e.

Constructing

maximum likelihood decoding function associated with a given group code

Theorem 2

If K is a finite subgroup of a group G, then every left coset of K in G has exactly as many elements as K.

Proof

Let aK be a left coset of K in G, where $a \in G$. Consider the function $f: K \to aK$ defined by

$f(k) = ak$, for $k \in K$.

We show that f is one to one and onto.

To show that f is one to one, we assume that

$f(k_1) = f(k_2)$, $k_1, k_2 \in K$.

Then $ak_1 = ak_2$

Left multiply a^{-1}, we have

$k_1 = k_2$

Proof

To show that f is onto, let b be an arbitrary element in aK. Then $b = ak$ for some $k \in K$. We now have

$f(k) = ak = b$

so f is onto.

Since f is one to one and onto, K and aK have the same number of elements.

Q.E.D

Group code word

Let e: $B^m \rightarrow B^n$ be an (m, n) encoding function that is a group code.

Thus the set N of code words in B^n is a subgroup of B^n whose order is 2^m, say

$N = \{x^{(1)}, x^{(2)}, \ldots, x^{(2^m)}\}$.

Coset leader – 陪集头

Suppose that the code word $x = e(b)$ is transmitted and that the word x_t is received.

The left coset of x_t is

$x_t \oplus N = \{\varepsilon_1, \varepsilon_2, \ldots, \varepsilon_{2m}\}$ where $\varepsilon_i = x_t \oplus x^{(i)}$

if ε_j is a coset member with smallest weight, then $x^{(j)}$ must be a code word that is closest to x_t.

An element ε_j , having smallest weight, is called a *coset leader*.

Procedure

For obtaining a maximum likelihood decoding function *d* associated with a given group code $e: B^n \to B^m$

Procedure

Step 1: Determine all the left cosets of $N = e(B^m)$ in B^n

Step 2: For each coset, find a coset leader (a word of least weight).

Step 3: If the word x_t is received, determine the coset of N to which x_t belongs.

Step 4: Let ε be a coset leader for the coset determined in Step 3. Compute $x = x_t \oplus \varepsilon$. If $x = e(b)$, we let $d(x_t) = b$. That is, we decode x_t as b.

Decoding table

Constructing a decoding table, each row is a left coset of *N* with the first element $\varepsilon^{(i)}$ the coset leader.

$\overline{0}$	$x^{(2)}$	$x^{(3)}$	$\cdots$	$x^{(2^m-1)}$
$\epsilon^{(2)}$	$\epsilon^{(2)} \oplus x^{(2)}$	$\epsilon^{(2)} \oplus x^{(3)}$	$\cdots$	$\epsilon^{(2)} \oplus x^{(2^m-1)}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$
$\epsilon^{(2^r)}$	$\epsilon^{(2^r)} \oplus x^{(2)}$	$\epsilon^{(2^r)} \oplus x^{(3)}$	$\cdots$	$\epsilon^{(2^r)} \oplus x^{(2^m-1)}$

Decoding table

If we receive the word x_t, we locate it in the table. If x is the element of N that is at the top of the column containing x_t, then x is the code word closest to x_t.

if $x = e(b)$, then $d(x_t) = b$.

$\overline{0}$	$x^{(2)}$	$x^{(3)}$	$\cdots$	$x^{(2^m-1)}$
$\epsilon^{(2)}$	$\epsilon^{(2)} \oplus x^{(2)}$	$\epsilon^{(2)} \oplus x^{(3)}$	$\cdots$	$\epsilon^{(2)} \oplus x^{(2^m-1)}$
$\vdots$	$\vdots$	$\vdots$		$\vdots$
$\epsilon^{(2^r)}$	$\epsilon^{(2^r)} \oplus x^{(2)}$	$\epsilon^{(2^r)} \oplus x^{(3)}$	$\cdots$	$\epsilon^{(2^r)} \oplus x^{(2^m-1)}$

Example 4

Consider the (3, 6) group code

N = {000000, 001100, 010011, 011111, 100101, 101001, 110110, 111010}

= $\{x^{(1)}, x^{(2)},\ldots, x^{(8)}\}$.

$$H=\begin{bmatrix}1&0&1\\0&1&1\\1&0&0\\1&0&0\\0&1&0\\0&0&1\end{bmatrix}\begin{bmatrix}0&0&0\\0&0&1\\0&1&0\\0&1&1\\1&0&0\\1&0&1\\1&1&0\\1&1&1\end{bmatrix}*\begin{bmatrix}1&0&0&1&0&1\\0&1&0&0&1&1\\0&0&1&1&0&0\end{bmatrix}=\begin{bmatrix}0&0&0&0&0&0\\0&0&1&1&0&0\\0&1&0&0&1&1\\0&1&1&1&1&1\\1&0&0&1&0&1\\1&0&1&0&0&1\\1&1&0&1&1&0\\1&1&1&0&1&0\end{bmatrix}$$

Example 4
Constructing decoding table:

000000	001100	010011	011111	100101	101001	110110	111010
000001	001101	010010	111110	100100	101000	110111	111011
000010	001110	010001	011101	100111	101011	110100	111000
000100	001000	010111	011011	100001	101101	110010	111110
010000	011100	000011	001111	110101	111001	100110	101010
100000	101100	110011	111111	000101	001001	010110	011010
000110	001010	010101	011001	100011	101111	110000	111100
010100	011000	000111	001011	110001	111101	100010	101110

001010	000110	011001	010101	101111	100011	111100	110000

Simplified decoding technique

Suppose that the (m, n) group code is e_H: $B^m \to B^n$, where **H** is a given parity check matrix.

$$\mathrm{H} = \begin{bmatrix} \mathrm{H}_{m\times r} \\ \mathrm{I}_r \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & \cdots & h_{1r} \\ h_{21} & h_{22} & \cdots & h_{2r} \\ \vdots & \vdots & & \vdots \\ h_{m1} & h_{m2} & \cdots & h_{mr} \\ \hline 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

Simplified decoding technique

Then the function $f_H: B^n \to B^r$ defined by

$f_H(x) = x * \mathbf{H}, \;\; x \in B^n$

is a homomorphism from the group B^n to the group B^r.

Theorem 3

If m, n, r, **H**, and f_H are as defined, then

f_H is onto.

Proof

Let $b = b_1 b_2 \ldots b_r$ be any element in B^r.

Letting $x = 0_1 \ldots 0_m b_1 b_2 \ldots b_r$

Then $x * \mathbf{H} = b$.

Thus $f_H(\mathrm{x}) = b$, so f_H is onto.

Q.E.D

Syndrome – 特征值

It follows from Corollary I of Section 9.5 that B^r and B^n/N are isomorphic, where

$N = \{x \in B^n \mid x*\mathbf{H} = \mathbf{0}\} = \ker(f_H) = e_H(B^m)$

under the isomorphism g: $B^n/N \to B^r$ defined by

$g(xN) = f_H(x) = x*\mathbf{H}$

The element $x*\mathbf{H}$ is called the *syndrome* of x

Theorem 4

Let x and y be elements in B^n. Then

x and y lie in the same left coset of N in B^n

if and only if

$f_H(x) = f_H(y)$

if and only if

they have the same syndrome.

Proof

It follows from Theorem 4 of Section 9.5 that x and y lie in the same left coset of N in B^n if and only if $x \oplus y = (-x) \oplus \mathrm{y} \in N$.

Since $N = \ker(f_H)$,

$x \oplus y \in \mathrm{N}$

if and only if

$f_H(x \oplus y) = 0_{Br}$

$f_H(x) \oplus f_H(\mathrm{y}) = 0_{Br}$

$f_H(x) = f_H(\mathrm{y})$

Q.E.D.

Decoding procedure

Suppose that we compute the syndrome of each coset leader. If the word x_t is received, we also compute $f_H(x_t)$, the syndrome of x_t. By comparing $f_H(x_t)$ and the syndromes of the coset leaders, we find the coset in which x_t lies. Suppose that a coset leader of this coset is ε. We now compute $x = x_t \oplus \varepsilon$. If $x = e(b)$, we then decode x_t as b.

New procedure

Step 1: Determine all left cosets of $N = e_H(B^m)$ in B^n.

Step 2: For each coset, find a coset leader, and compute the syndrome of each leader

Step 3: If x_t is received, compute the syndrome of x_t and find the coset leader ε having the same syndrome. Then $x_t \oplus \varepsilon = x$ is a code word $e_H(b)$, and $d(x_t) = b$.

Example 5

Consider the parity check matrix and the (3, 6) group code e_H: $B^3 \rightarrow B^6$.

$$H = \begin{bmatrix} 110 \\ 101 \\ 011 \\ 100 \\ 010 \\ 001 \end{bmatrix} \qquad \left.\begin{array}{l} e(000) = 000000 \\ e(001) = 001011 \\ e(010) = 010101 \\ e(011) = 011110 \\ \hline e(100) = 100110 \\ e(101) = 101101 \\ e(110) = 110011 \\ e(111) = 111000 \end{array}\right\} \text{code word}$$

Example 5

N = {000000, 001011, 010101, 011110, 100110, 101101, 110011, 111000}

Syndrome of Coset Leader	Coset leader
000	000000
001	000001
010	000010
011	001000
100	000100
101	010000
110	100000
111	001100

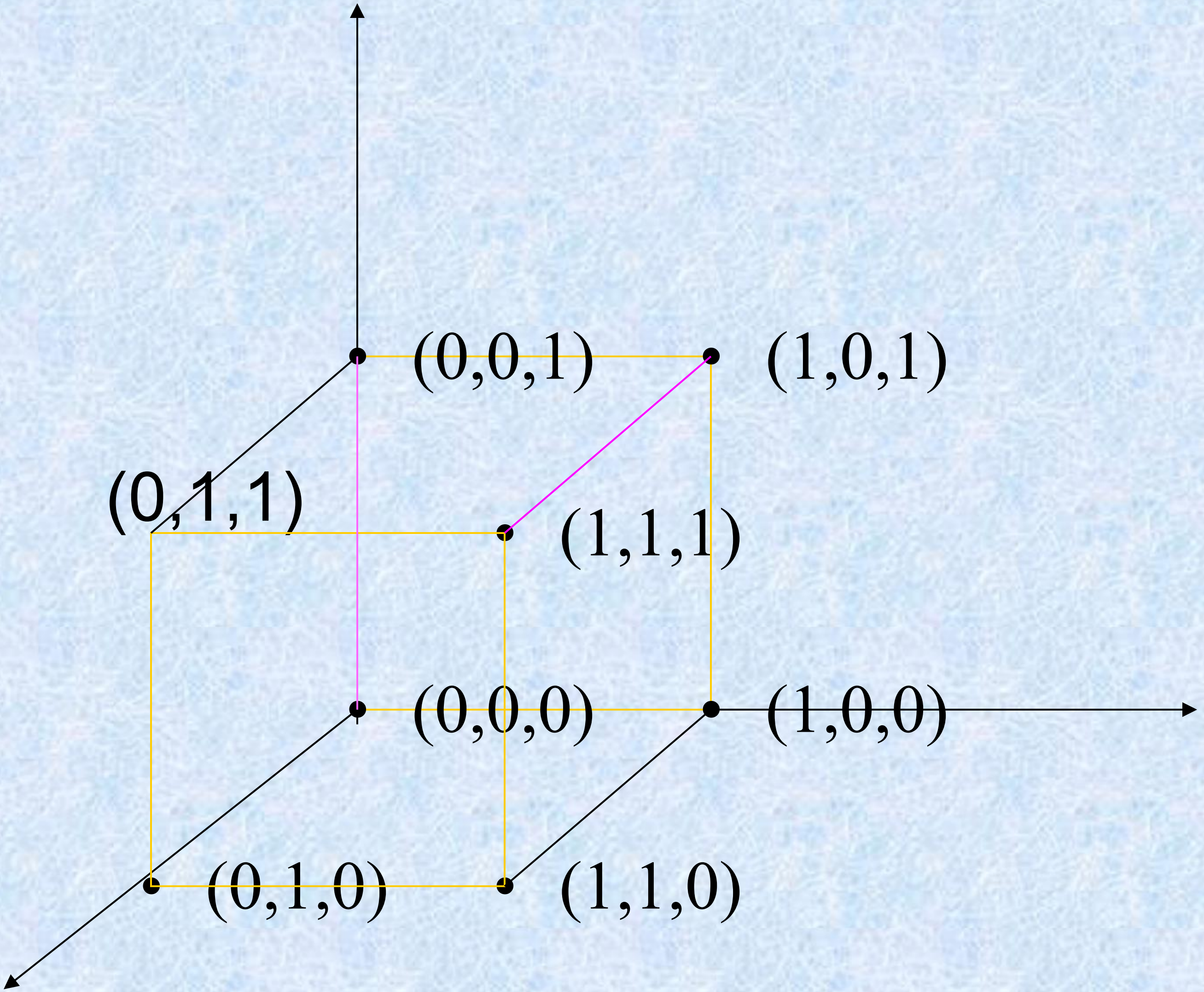
Example 5

If x_t = 001110, then $f_H(x_t) = x_t$ * **H** = 101, same as ε=010000.

$x = x_t \oplus \varepsilon$ = 001110 $\oplus$ 01000 = 011110 = e(011), so decode 001110 as 011.

Syndrome of Coset Leader	Coset leader
000	000000
001	000001
010	000010
011	001000
100	000100
101	010000
110	100000
111	001100

$B^m(0,1) \rightarrow B^n(000,111)$



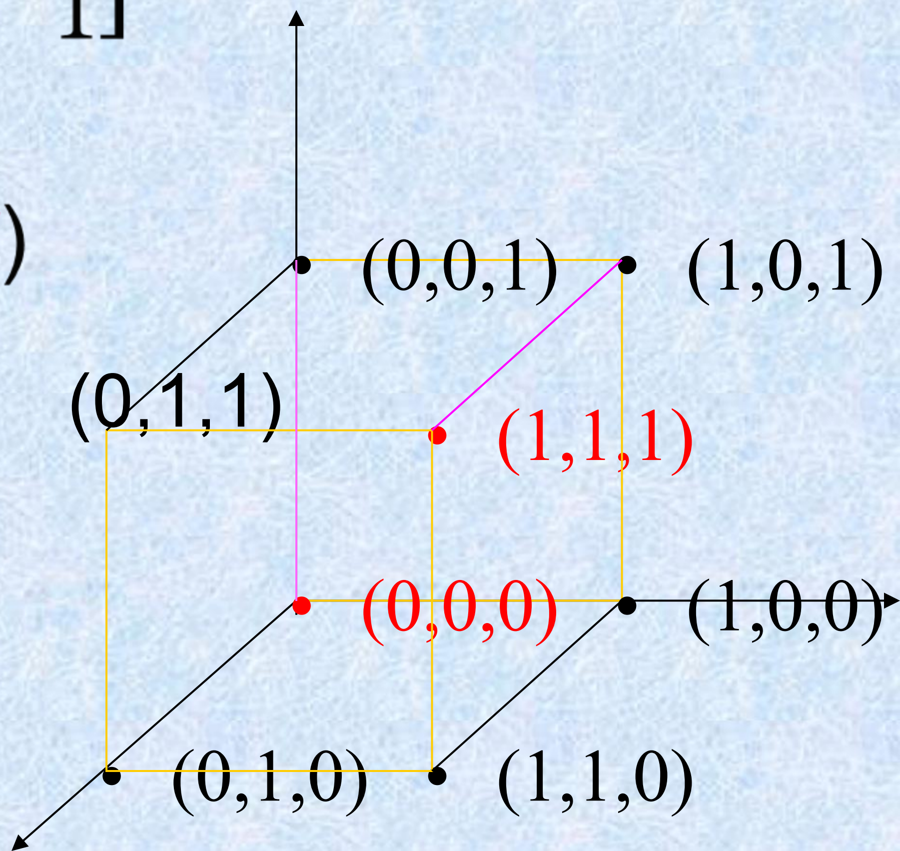
编码：$\begin{bmatrix}0\\1\end{bmatrix} * \begin{bmatrix}1 & 1 & 1\end{bmatrix} = \begin{bmatrix}0 & 0 & 0\\1 & 1 & 1\end{bmatrix}$ 码字 $B^1 \to B^3$

$B^m \to B^n$

一致性校验矩阵

$H=\begin{bmatrix}1 & 1\\1 & 0\\0 & 1\end{bmatrix}$ $\begin{bmatrix}0 & 0 & 0\\1 & 1 & 1\end{bmatrix} * \begin{bmatrix}1 & 1\\1 & 0\\0 & 1\end{bmatrix} = \begin{bmatrix}0 & 0\\0 & 0\end{bmatrix}$ $x*H=0$

译码表 Coset leader $B^2(B^r)$



Homework

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Key ideas for review

Message, word

(m, n) encoding function, one-to-one

- *Code word, parity check code*
- *Detect, correct, k or fewer errors*

Hamming distance

- *Properties of distance*

Group code and parity check matrix

- *Minimum distance of a group code*

Maximum likelihood technique

- *Maximum likelihood decoding function*
- *Syndrome and decoding procedure for group code*